

Homemade Motor Fuel Through Wood Gasification

By **The Mother Earth News Editors**

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WoodPEC's six-cylinder Ford uses a wood gasification process to burn firewood for fuel.

PHOTO: MOTHER EARTH NEWS STAFF

As we're sure everyone realizes, MOTHER EARTH NEWS doesn't claim to have a corner on innovation. So when given an opportunity to show off someone else's ingenuity, we usually take it and hope that—by doing so—we provide our readers with new concepts which they can put to use in developing their own projects.



Recently, Morton Fry (of the Miles W. Fry & Son nursery) contacted our North Carolina home office and explained that he and a group of associates were working on a wood-powered energy source in which he thought we might be interested. And, since our researchers had been experimenting with wood gasification units too, we took Mort up on his offer and paid a visit to the Fry family headquarters in Pennsylvania's rural Lancaster County.

Mr. Fry and about a dozen other concerned people have pooled their resources and talent to form the Wood Power Energy Corporation, a group that's dedicated to making the ligneous combustible a primary source of fuel in this nation by the turn of the century. Rather than merely promote the use of wood in its *natural* form, "WoodPEC" hopes to be instrumental in the development of some new and more *efficient* cellulose-based energy systems.

Working under a DOE/General Electric grant, one of the association's members—University of Pennsylvania biochemist Dr. E. Kendall Pye—has *already* been successful in converting tree-derived polysaccharide material into fermentable glucose, used for making ethanol. And the group eventually plans to get involved not only in the improvement of wood-fired steam generators *and* the turbines which operate from them, but also in the conversion of "timber" carbons into conventional liquid fuels, using a heat-and-catalyst process.

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But for the present the big news at the Frys' Ephrata, Pennsylvania tree farm is the woodburning pickup truck that the WoodPEC folks have developed. The gasification *concept* used in the vehicle is, for the most part, the same as that utilized in any of the producer-gas generators pressed into service during the fuel-lean years of World War II. The Keystone Staters have, however, apparently improved on the *design* of the gas-producing unit itself, thus rendering it [1] more efficient than the wood-vapor generators utilized 40 years ago, and [2] more convenient, since it doesn't have to be maintained as diligently as did the original units.

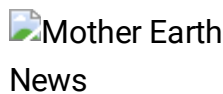
Because the WoodPEC people feel that they've hit upon a truly unique arrangement, they're in the process of patenting their design, so we've agreed not to reveal any "privileged" information at this time. We can, however, say that the 14-year-old Ford on which the gasifier's been installed doesn't run much differently from any *conventionally* powered vehicle with 200,000 miles on its "clock." The well-used pickup has been modified to operate on producer gas alone (with no dual fuel capability) and has been driven a total of 800 miles in that mode. Start-up time (that period between lighting the fire and actually driving away) is about ten minutes, and the rolling workhorse *can* attain speeds over the legal limit.

Though "timber power" admittedly isn't as *convenient* as is gasoline, it's a whole lot less expensive, since even *scrap* wood can be burned. Typical fuel economy figures work out to about one mile per pound. Additionally—because they've been able to achieve higher-than-normal temperatures within the "hearth" zone of their wood gasifier—the "WoodPECers" find that they're extracting as much energy as possible from their fuel while realizing the additional benefit of minimal ash accumulation in the firebox. (Morton Fry estimates that two or three

Now, with their transportation experiment for the most part completed, the energy-conscious researchers are busy investigating a new project: using producer gas to power a stationary generator which will, they hope, not only provide electricity for part of the firm's nursery complex during periods of peak demand, but—in addition—feed energy *back* into the grid during “slack” spells. Meanwhile, the folks here at y(our) favorite magazine are working along *similar* lines, and want to incorporate a few new twists besides. So keep watching, over the next few months, for fresh developments!



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